DURABILITY OF AR GLASS FIBER REINFORCED PLASTIC BARS

by Fares E. Tannous[1] and Hamid Saadatmanesh[2]

ABSTRACT: Since the development of reinforced concrete over a century ago, engineers have constantly struggled with the corrosion problem of steel reinforcement. The new advances in the field of materials science have resulted in the development of fiber reinforced plastic (FRP) materials that could potentially be used as bars in reinforced concrete structures. The attractive features of this type of bars include high strength, light weight, and resistance to electrochemical corrosion. E-glass is the most widely used fiber material in FRPs; however, studies have shown that this type of fiber is susceptible to degradation in alkaline environments (i.e., concrete). This paper presents the results of a durability study on FRP bars made from an alternative glass fiber, alkali resistant (AR) glass, that could potentially improve the durability of FRP bars. A total of 160 bar samples were tested after exposure to corrosive solutions at temperatures of 25°C and 60°C. These solutions were designed to simulate accelerated exposure to adverse conditions in the field. Test variables included two matrix materials (polyester and vinyl ester), seven chemical solutions, and ultraviolet radiation. Moisture absorption by bars and associated changes in the mechanical properties were the focus of this study. In addition, ten reinforced concrete beams were tested in flexure until failure, to examine the effects of exposure to concrete and deicing salts. Test results indicated that significant loss of strength could result from exposure of AR glass FRP bars to simulated aggressive field conditions. An analytical study based on diffusion models and Fick's law was also conducted to predict the loss of strength. In the Fickian range of diffusion, the models predicted the loss of strength with a reasonable accuracy.

INTRODUCTION

Corrosion of steel reinforcement in harsh environments such as in coastal regions or in bridges where deicing salts are used is one of the major infrastructure problems facing civil engineers (Gibson 1987; "Our" 1990). The maintenance cost to prevent corrosion of steel reinforcement, and the high cost of repair or replacement of concrete structures that are deteriorated from the corrosion induced damage, have evoked a worldwide interest in alternative materials for reinforcing concrete structures.

Fiber reinforced plastics (FRPs) have been increasingly gaining the interest of researchers and engineers as attractive alternatives to steel due to their unique properties such as high strength-to-weight ratio, resistance to electrochemical corrosion, and magnetic neutrality. E-glass fiber composites are the most widely used because of their lower cost. Studies, however, have shown serious durability problems in FRP bars made from E-glass in environments with high alkalinity such as in concrete (Tannous 1997; Katsuki and Uomoto 1995). An alternative glass fiber that could potentially improve the durability of FRPs in this type of environment is the alkali resistant (AR) glass (*AGR* 1994). Currently, AR glass fibers are used for various purposes in civil engineering. For example, short AR glass fibers have been used in the construction of wall panels for high-rise buildings in the United States because of their light weight and durability (Sen et al. 1993). Before they can be used as reinforcement for concrete structures, however, the durability of FRP bars made from AR glass fibers in civil engineering environments must be investigated.

The primary objective of this study is to examine the durability of AR glass FRP bars in simulated aggressive field conditions. The durability of these bars is examined through changes in their ultimate tensile strength (P_u), ultimate strain (ε_u), and elastic modulus (E), after exposure to various chemical solutions simulating accelerated exposure to field conditions. Durability studies performed on E-glass FRP bars have shown a significant loss of strength due to exposure to alkaline environments (Uomoto and Katsuki 1994; Tannous 1997). The hydroxide ions OH^- in an alkaline environment attack the primary component of glass (silica or SiO_2) and cause a breakdown in the Si-O-Si single bond forming the glass molecular structure. This results in fiber corrosion and loss of strength. The high zirconia (ZrO_2) content of AR-glass could potentially improve its behavior in this type of environment.

In this study, a comprehensive experimental and analytical investigation was undertaken to examine the durability of AR glass FRP bars in civil engineering environments. Eight different environments were simulated in the direct exposure tests of the bars. In addition, 10 beam specimens reinforced with AR glass FRP bars were exposed to deicing salt solutions for a period of two years and then tested in flexure. The analytical study was based on the diffusion models and Fick's law. These models were used to predict the depth from the surface of bar where the ions had penetrated. The penetrated zone was assumed ineffective for carrying the tensile load of the bar since the fibers that are the primary load carrying component of the composite bar were assumed damaged in that zone. The residual strength of the bar was then predicted based on the unpenetrated zone of the cross section.

BACKGROUND

The increasing use of fiber reinforced composites in civil engineering applications requires a thorough understanding of the durability of such materials when exposed to harsh environments. To accomplish this task, long-term durability studies of these materials in aggressive field conditions must be conducted. A primary cause of deterioration of FRPs is the diffusion of moisture and other corrosive solutions into the matrix, which can damage the matrix as well as the fibers. Therefore, moisture absorption and associated changes in material properties must be examined. Moisture absorption of a composite can be defined in terms of two parameters: maximum moisture content or saturation moisture (M_m), and mass diffusion coefficient (D). Maximum moisture content is the moisture level in the composite that is reached asymptotically after a long period of time. M_m is dependent on material type and temperature and type of the environment. Mass diffusion

[1]Staff Struct. Engr., Vanasse Hangen Brustlin Inc., Providence, RI; formerly, Grad. Res. Asst. Dept. of Civ. Engrg. and Engrg. Mech., Univ. of Arizona, Tucson, AZ 85721.

[2]Prof., Dept. of Civ. Engrg. and Engrg. Mech., Univ. of Arizona, Tucson, AZ.

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J. Compos. Constr., 1999, 3(1): 12-19

coefficient, or diffusivity, is a measure of the speed at which the moisture is transported through the material. The diffusivity is dependent on the material type, moisture content within the composite, and the type and temperature of the environment.

Few studies have been conducted to investigate the durability of glass fiber reinforce plastic (GFRP) bars in solutions that simulated exposure to civil engineering environments. Katsuki and Uomoto (1995) have determined that short-term moisture diffusion into GFRP rods was Fickian. GFRP rods were immersed in 1.0 mol/L NaOH solution with pH of 13 at 40°C. The recorded diffusivity was 2.8×10^{-6} cm^2/h (4.34×10^{-7} sq in./h). Losses in strength of up to 60% were recorded after 120 days.

In another study, the durability of 10 mm (3/8 in.) diameter E-glass FRP bars in a marine environment was examined (Zayed 1991). Samples were placed in saturated calcium hydroxide solution, NaCl 3.5% by weight solution, and water at temperatures of 25 and 40°C. Losses in strength of up to 15% were reported in alkaline solution after 85 days. The strength loss for samples in water at 40°C was reported at 5%. Initial weight gain followed Fickian diffusion, but for longer periods of time it became non-Fickian.

EXPERIMENTAL PROGRAM

A total of 160 bar samples and 10 concrete beams were tested to examine the durability of AR glass bars. In addition to the bar specimens that were exposed to different chemical solutions and then tested in tension to determine the loss of mechanical properties, a number of shorter bar segments [100 mm (4 in.) long] were also exposed to the same environments to determine the diffusivity coefficient D. Two type of matrices were used in the construction of bars, namely, polyester and vinyl ester. The specimens were exposed to eight different chemical solutions, each simulating a specific field condition. The following sections briefly describe the materials, and specimen preparation and testing.

Materials

The bars used in the study were pultruded from AR glass fibers in polyester or vinyl ester resin matrix. The fibers were passed through a resin bath and pulled through a die of selected size. Two bar diameters of 10 mm (3/8 in.) and 19.5 mm (3/4 in.) were produced. According to the manufacturer, the fiber volume fraction v_f was 72%. To improve the mechanical bond to concrete, glass fibers were also wound around the bar to form a ribbed surface. Fig. 1 shows typical samples of these bars.

FIG. 1. Samples of E-Glass and AR Glass FRP Bars

Tension Test Specimens

Bar samples used in the tension tests were 800 mm (32 in.) long and were instrumented with CEA-06-125UN-120 electrical resistance strain gauges with a resolution of 1×10^{-6} in./in. All samples were tested in tension to failure at room temperature and the stress-strain relationship for each sample was recorded. The load was applied at the rate of 1.27 mm/min (0.05 in./min) until failure. The ends of the samples were coated with a moisture and chemical resistant epoxy to reduce diffusion through the ends. Samples were oven treated at a temperature of 65°C until no change in weight was observed (i.e., initial moisture content was zero), before they were placed in various solutions.

Tests for Diffusion Coefficient *D*

The specimens used to determine the saturation moisture content M_m and diffusivity D for the 10 mm (3/8 in.) and 19.5 mm (3/4 in.) diameter AR glass FRP bars were 102 mm (4 in.) long bar segments. As for tension specimens, the ends of these bars were also capped with a moisture and chemical resistant epoxy, and they were oven treated before they were placed in different solutions. Weight changes were recorded on a Mettler analytical balance with a 10^{-4} gram resolution. The objective of these tests was to determine the percent weight gain of the composite material as a function of the square root of time for determining the diffusion coefficient D. The smaller size of these specimens allowed a more accurate measurement for the weight gain for determining the saturation moisture content M_m and diffusion coefficient D.

Simulated Aggressive Environments (Solutions)

Eight different environments were simulated for accelerated exposure of specimens to field conditions. These were (1) water (H_2O) at 25°C, simulating exposure to high humidity; (2) saturated $Ca(OH)_2$ solution with pH of 12 at 25°C, simulating exposure to hydrating cement; (3) saturated $Ca(OH)_2$ solution with pH of 12 at 60°C, simulating exposure to hydrating cement at higher temperature; (4) HCl solution with pH of 3 at 25°C, simulating exposure to acidic environment; (5) NaCl 3.5% by weight solution at 25°C, simulating exposure to seawater (Mehta and Haynes 1975); (6) NaCl + $CaCl_2$ (2:1) 7% by weight solution at 25°C, simulating exposure to sodium chloride and TETRA94 deicing salts (''Calcium'' 1992); (7) NaCl + $MgCl_2$ (2:1) 7% by weight solution at 25°C, simulating exposure to sodium chloride and Ice-Stop deicing salts (''Ice-Stop'' 1992); and (8) ultraviolet radiation at the rate of 31.7×10^{-6} J/s/cm^2 (2.04×10^{-4} J/s/sq in.), simulating exposure to sunlight in a desert environment for the 10 mm (3/8 in.) bars only. Further details on the solution preparation are given in a comprehensive report (Tannous 1997).

Beam Specimens

A total of 10 concrete beams measuring 2.4 m (8 ft) in length with a 200×405 mm (8×16 in.) rectangular cross section were tested in flexure until failure. Each beam was reinforced with two 10 mm (3/8 in.) diameter AR glass/polyester or vinyl ester bars, resulting in a reinforcement ratio of 0.18%, as shown in Fig. 2. Shear reinforcement consisted of 10 mm (3/8 in.) diameter double-leg steel stirrups placed at 125 mm (5 in.) on center. The average concrete compressive strength was 34.82 MPa (5.05 ksi). All beams were tested in four point bending.

MOISTURE DIFFUSION

Diffusion Models

The single free phase diffusion model, i.e., Fick's law (Tsai and Hahn 1985), and the two-phase Langmuir model (Carter

J. Compos. Constr., 1999, 3(1): 12-19

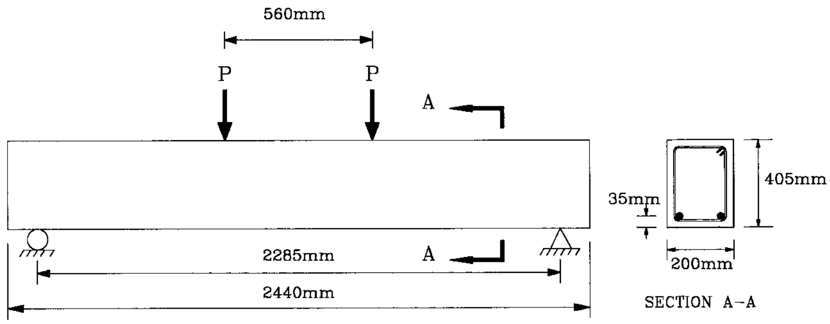

FIG. 2. Details of Test Beams

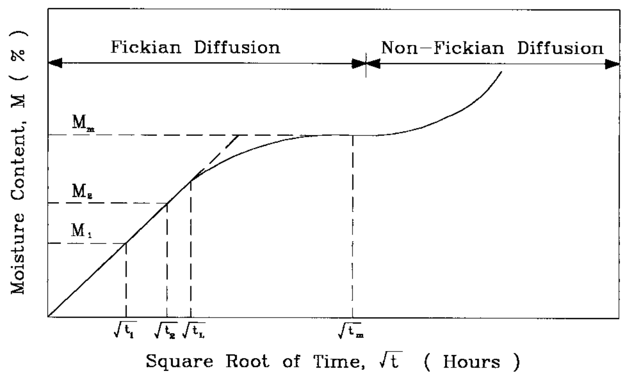

FIG. 3. Typical *M* (%) versus $t^{1/2}$ Relationship for FRP Bars

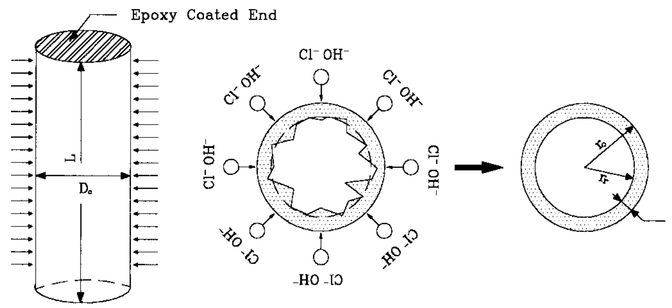

FIG. 4. Diffusion of Cl^- and OH^- Free Ions from Alkaline, Acidic, and Salt Solutions

and Kibler 1978; Gurtin and Yatomi 1979; Lo et al. 1982) were used in the analytical study to describe the diffusion process in the FRP bars. These models were used to determine the diffusion coefficient D from the plots of moisture content versus square root of time, as shown in Fig. 3. Complete details of these models as well as the procedures for determining D are given by Tannous (1997).

Loss of Strength

The durability of AR glass FRP bars is examined in terms of changes in the ultimate tensile strength P_u, elastic modulus E, and ultimate strain at failure ε_{uf}. The loss in strength is determined based on the following two assumptions: (1) The glass fibers in the penetrated area are damaged and no longer carry load; and (2) the fibers and matrix in the intact zone are undamaged and possess the same initial mechanical properties.

The alkali and salt attack will mainly occur due to the presence of OH^- and Cl^- free ions in the solution. Other ions present in the solution could to a lesser degree react with the fiber and the matrix. In this study, only the effects of OH^- and Cl^- ions were investigated. The Cl^- ions are not as damaging to glass as is the OH^- ion, however Cl^- ions can penetrate the matrix, causing microcracking and fracturing of the matrix and resulting in rapid moisture diffusion as well as debonding of the fibers. Debonding of the fibers, in turn, will result in the loss of strength of the bar.

The loss in strength is predicted based on the depth of moisture penetration as shown in Fig. 4. The depth of penetration x (mm) is calculated from (Katsuki and Uomoto 1995)

$$x = \sqrt{2DCt} \tag{1}$$

where D = diffusion coefficient (mm²/min), as determined from the plots of moisture content versus square root of time (Tannous 1997); C = normalized or relative term describing the alkaline or chloride concentration [(mol/L)/(mol/L)]; and t = elapsed time (min). Considering the initial radius of the bar to be r_0 (mm), the radius of the residual area to be r_r (mm), and the initial tensile strength to be P_0 (kN), then the predicted residual tensile strength P_p (kN) can be calculated from

$$P_p = P_u\left(1 - \frac{x}{r_0}\right)^2 \equiv P_u\left(\frac{r_r}{r_0}\right)^2 \tag{2}$$

RESULTS

Four different types of AR glass bars were tested: 10 mm (3/8 in.) AR glass/polyester; 10 mm (3/8 in.) AR glass/vinyl ester; 19.5 mm (3/4 in.) AR glass polyester; and 19.5 mm (3/4 in.) AR glass/vinyl ester. Each bar was designated by one numeral followed by three letters. The numeral represents the bar diameter in millimeters rounded to the nearest integer, 10 for the 10 mm bars and 20 for the 19.5 mm bars. The letters following the numeral are ARP for AR glass/polyester and ARV for AR glass/vinyl ester. For example, 10ARV means 10 mm (3/8 in.) diameter AR glass/vinyl ester bar.

The test samples were immersed in chemical solutions simulating different environments, and changes in weight were recorded periodically. The maximum or saturation moisture content M_m and diffusivity D of 10ARP, 10ARV, 20ARP, and 20ARV bars are given in Table 1. The percent weight gain M is based on the numerical average of the percentage weight gain of three identical samples. The initial material properties of rebars are listed in Table 2. These properties were obtained as the average test results of five samples successfully tested in tension. The test was considered successful only when the sample failed outside the grip area. The test results of samples that failed in the grip area were not considered. Similarly, the same number of samples was used to determine the material properties after six months of exposure to chemical solutions and ultraviolet radiation. Tension test results of 10ARP and 10ARV rebars after six months of exposure are listed in Table 3. Similar results are listed in Table 4 for the 20ARP and 20ARV rebars.

In all cases, the initial part of the M (%) versus $t^{1/2}$ relationship was approximately linear, indicating Fickian diffusion, then leveling off towards an asymptotic value of M_m. The non-Fickian diffusion, resulting primarily from matrix cracking, was indicated by the rapid nonlinear increase in M, at which the one-phase and two-phase diffusion models became invalid for the prediction of the behavior. Generally, the one-phase and two-phase model predictions were very close.

J. Compos. Constr., 1999, 3(1): 12-19

TABLE 1. Maximum Moisture Content (M_m) and Diffusivity (D) of 10 mm (3/8 in.) and 19.5 mm (3/4 in.) Diameter AR Glass/Polyester and Vinyl Ester Rebars Immersed in Solutions

		10 mm ϕ AR Glass Polyester Matrix		10 mm ϕ AR Glass Vinyl Ester Matrix		19.5 mm ϕ AR Glass Polyester Matrix		19.5 mm ϕ AR Glass Vinyl Ester Matrix	
Environment (1)	T (°C) (2)	M_m (%) (3)	$D \times 10^7$ (mm²/min) (4)	M_m (%) (5)	$D \times 10^7$ (mm²/min) (6)	M_m (%) (7)	$D \times 10^7$ (mm²/min) (8)	M_m (%) (9)	$D \times 10^7$ (mm³/min) (10)
Water (H_2O)	25°C	1.650	4.20	0.724	2.10	1.214	5.20	0.533	2.60
$Ca(OH)_2$, pH = 12	25°C	1.401	3.10	0.527	1.20	0.631	4.50	0.857	2.30
$Ca(OH)_2$, pH = 12	60°C	1.528	5.10	0.631	1.70	0.745	9.80	1.026	3.10
HCl, pH = 3	25°C	1.023	6.40	0.510	3.90	0.716	6.30	0.512	2.90
NaCl, 3.5%	25°C	0.603	4.20	0.436	1.10	0.552	9.20	0.605	5.30
NaCl + $CaCl_2$ (2:1), 7%	25°C	0.635	4.10	0.651	2.50	0.671	10.7	0.613	5.90
NaCl + $MgCl_2$ (2:1), 7%	25°C	0.581	4.40	0.686	2.70	0.695	11.3	0.715	5.60

TABLE 2. Initial Material Properties of 10 mm (3/8 in.) and 19.5 mm (3/4 in.) Diameter AR Glass/Polyester and Vinyl Ester FRP Rebars

Material property (1)	10 mm ϕ AR glass polyester matrix (2)	10 mm ϕ AR glass vinyl ester matrix (3)	19.5 mm ϕ AR glass polyester matrix (4)	19.5 mm ϕ AR glass vinyl ester matrix (5)
D_0 (mm)	10.434	10.158	19.554	19.352
A_g (mm²)	85.51	81.04	270.4	294.2
v_f (%)	72	72	72	72
P_u (kN)	60.00	61.29	171.3	176.1
SD (kN)	3.976	4.851	10.322	7.988
σ_u (MPa)	701.7	756.3	633.5	598.6
SD (MPa)	46.5	59.86	38.17	27.15
E (GPa)	53.8	54.1	53.9	51.2
SD (GPa)	5.83	5.78	5.46	5.01
ε_u (%)	1.304	1.398	1.175	1.169
SD (%)	0.137	0.162	0.181	0.204

However, the predicted results obtained from the two-phase model were more accurate in the transitional zone between the end of the linear part and the part asymptotic to M_m. Both models converged towards the same asymptotic M_m value.

Changes in the elastic modulus were generally small. This indicates that the damage caused to fibers and matrix due to moisture diffusion was confined to the penetrated zone, while the remainder zone possessed the same initial material properties. In all cases, the ultimate strain at failure was lower than the initial strain, indicating a more brittle behavior after exposure as was also indicated by the increase in the elastic modulus. A brief description of the behavior of bars exposed to different environments for a period of six months is given in the following.

Water (H_2O): T = 25°C

For bar samples immersed in pure water, M_m and D were higher for polyester rebars than for vinyl ester. For example,

TABLE 3. Test Results of 10 mm (3/8 in.) Diameter AR Glass/Polyester and Vinyl Ester Rebars after Six Months of Exposure

Environment (1)	$D \times 10^7$ (mm²/min) (2)	C [(mol/L)/(mol/L)] (3)	$t \times 10^5$ (min) (4)	x (mm) (5)	P_u (kN) (6)	P_r (kN) (7)	P_p (kN) (8)	E (GPa) (9)	E_r (GPa) (10)	ε_u (%) (11)	ε_{ur} (%) (12)
(*a*) 10 mm ϕ AR glass/polyester											
Water (H_2O), T = 25°C	4.2	0	2.592	—[a]	60.0	55.6	—[a]	53.8	52.6	1.304	0.952
$Ca(OH)_2$ solution, pH = 12, T = 25°C	3.1	1.041	2.592	0.411	60.0	47.5	50.9	53.8	51.7	1.304	1.082
$Ca(OH)_2$ solution, pH = 12, T = 60°C	5.5	1.041	2.592	0.525	60.0	43.2	48.1	53.8	53.9	1.304	0.948
HCl solution, pH = 3, T = 25°C	6.4	0.001	2.592	0.018	60.0	52.8	59.6	53.8	54.1	1.304	1.141
NaCl 3.5% by weight solution	4.2	0.600	2.592	0.361	60.0	50.1	52.0	53.8	53.3	1.304	1.104
NaCl + $CaCl_2$ (2:1), 7% by weight solution	4.1	1.218	2.592	0.509	60.0	45.6	48.9	53.8	52.2	1.304	1.032
NaCl + $MgCl_2$ (2:1), 7% by weight solution	4.4	0.900	2.592	0.453	60.0	44.3	50.0	53.8	55.1	1.304	0.948
Ultraviolet radiation (31.7×10^{-6} J/s/cm²)	—[a]	—[a]	2.592	—[a]	60.0	56.9	—[a]	53.8	52.1	1.304	1.185
(*b*) 10 mm ϕ AR glass/vinyl ester											
Water (H_2O), T = 25°C	2.1	0	2.592	—[a]	61.3	57.7	—[a]	54.1	56.3	1.398	1.159
$Ca(OH)_2$ solution, pH = 12, T = 25°C	1.2	1.041	2.592	0.255	61.3	53.5	55.3	54.1	54.4	1.398	1.215
$Ca(OH)_2$ solution, pH = 12, T = 60°C	1.7	1.041	2.592	0.303	61.3	47.4	54.2	54.1	53.3	1.398	1.102
HCl solution, pH = 3, T = 25°C	3.9	0.001	2.592	0.014	61.3	53.2	61.0	54.1	58.2	1.398	1.128
NaCl 3.5% by weight solution	1.1	0.600	2.592	0.181	61.3	52.3	57.0	54.1	56.7	1.398	1.140
NaCl + $CaCl_2$ (2:1), 7% by weight solution	2.5	1.218	2.592	0.397	61.3	47.4	52.1	54.1	57.3	1.398	1.026
NaCl + $MgCl_2$ (2:1), 7% by weight solution	2.7	0.900	2.592	0.355	61.3	45.9	53.0	54.1	59.1	1.398	0.964
Ultraviolet radiation (31.7×10^{-6} J/s/cm²)	—[a]	—[a]	2.592	—[a]	61.3	57.8	—[a]	54.1	55.4	1.398	1.079

[a]Fick's law was not used.

J. Compos. Constr., 1999, 3(1): 12-19

TABLE 4. Test Results of 19.5 mm (3/4 in.) Diameter AR Glass/Polyester and Vinyl Ester Rebars after Six Months of Exposure

Environment (1)	$D \times 10^7$ (mm^2/min) (2)	C [(mol/L)/(mol/L)] (3)	$t \times 10^5$ (min) (4)	x (mm) (5)	P_u (kN) (6)	P_r (kN) (7)	P_p (kN) (8)	E (GPa) (9)	E_r (GPa) (10)	ε_u (%) (11)	ε_{ur} (%) (12)
(*a*) 19.5 mm φ AR glass/polyester											
Water (H_2O), T = 25°C	5.2	0	2.592	—[a]	171.3	160.3	—[a]	53.9	51.4	1.175	1.034
$Ca(OH)_2$ solution, pH = 12, T = 25°C	4.5	1.041	2.592	0.493	171.3	148.5	153.5	53.9	56.4	1.175	0.977
$Ca(OH)_2$ solution, pH = 12, T = 60°C	9.8	1.041	2.592	0.727	171.3	142.3	145.5	53.9	54.3	1.175	0.975
HCl solution, pH = 3, T = 25°C	6.3	0.001	2.592	0.018	171.3	153.5	170.6	53.9	52.8	1.175	1.075
NaCl 3.5% by weight solution	9.2	0.600	2.592	0.535	171.3	159.7	152.1	53.9	58.2	1.175	1.017
NaCl + $CaCl_2$ (2:1), 7% by weight solution	10.7	1.218	2.592	0.822	171.3	154.2	142.3	53.9	55.6	1.175	1.032
NaCl + $MgCl_2$ (2:1), 7% by weight solution	11.3	0.900	2.592	0.726	171.3	155.7	145.5	53.9	57.3	1.175	1.010
Ultraviolet radiation (31.7 $\times 10^{-6}$ J/s/cm^2)	—[a]	—[a]	2.592	—[a]	171.3	—[a]	—[a]	53.9	—[a]	1.175	—[a]
(*b*) 19.5 mm φ AR glass/vinyl ester											
Water (H_2O), T = 25°C	2.6	0	2.592	—[a]	176.1	165.7	—[a]	51.2	53.6	1.169	1.152
$Ca(OH)_2$ solution, pH = 12, T = 25°C	2.3	1.041	2.592	0.352	176.1	158.7	163.6	51.2	52.3	1.169	1.032
$Ca(OH)_2$ solution, pH = 12, T = 60°C	3.1	1.041	2.592	0.409	176.1	151.2	161.6	51.2	50.2	1.169	1.025
HCl solution, pH = 3, T = 25°C	2.9	0.001	2.592	0.012	176.1	164.1	175.7	51.2	52.6	1.169	1.061
NaCl 3.5% by weight solution	5.3	0.600	2.592	0.406	176.1	165.2	161.7	51.2	54.7	1.169	1.023
NaCl + $CaCl_2$ (2:1), 7% by weight solution	5.9	1.218	2.592	0.610	176.1	160.0	154.6	51.2	52.6	1.169	1.038
NaCl + $MgCl_2$ (2:1), 7% by weight solution	5.6	0.900	2.592	0.511	176.1	161.1	158.0	51.2	55.4	1.169	0.991

[a]Fick's law was not used.

for 10ARP bars M_m and D were 1.65% and 4.2 $\times 10^{-7}$ mm^2/min (6.51 $\times 10^{-10}$ sq in./min), respectively, while M_m and D were 0.72% and 2.1 $\times 10^{-7}$ mm^2/min (3.26 $\times 10^{-10}$ sq in./min), respectively, for the 10ARV bar. For pure water, Fick's law was applicable in calculating the diffusivity, but it could not be used to calculate the depth of penetration since this equation applies to only cases where ions are present in the solution. Since the ion concentration C is equal to zero for pure water, (1) results in a trivial solution and is therefore not applicable for predicting the depth of penetration. The submersion in water resulted in a measured reduction in tensile strength of 7.3% and 5.9% for the 10ARP and 10ARV bars, respectively. Limited changes in the elastic modulus were observed, and these changes were within 5% of the initial values. Similar observations were made for the 20ARP and 20ARV rebars.

Alkaline Solution $Ca(OH)_2$: pH = 12; T = 25°C

Figs. 5 and 6 show the M (%) versus $t^{1/2}$ relationship for 10ARP and 10ARV bars in alkaline solution at 25°C. M_m and D were generally lower for the vinyl ester bars, which indicates that the particular vinyl ester used for the construction of the bars in this study provided a better protection than polyester against moisture diffusion. This is also reflected in a lower depth of penetration x, and thus a smaller predicted loss in tensile strength. The loss in strength was 20.8% for 10ARP bars, while it was 12.7% for 10 ARV bars. Figs. 7 and 8 show the M (%) versus $t^{1/2}$ relationship for 20ARP and 20ARV bars. Again, the diffusivity for vinyl ester was lower than that of polyester. For polyester bars, the average loss of strength was 13.3%, while for vinyl ester bars it was 9.9%. The percentage difference between the measured residual strength P_r and the predicted residual strength P_p was 3.4% in the case of the 20ARP bar, and

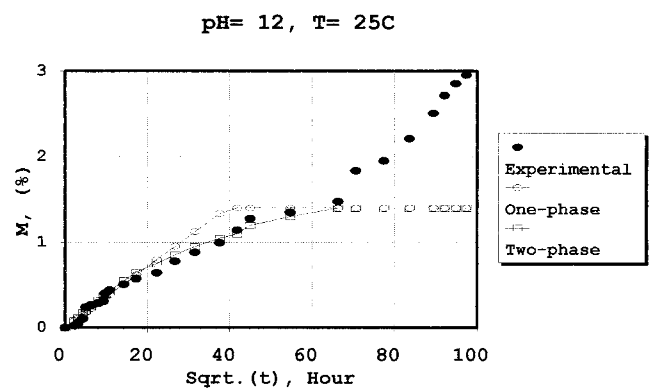

FIG. 5. M (%) versus $t^{1/2}$ for 10ARP Bar Specimen

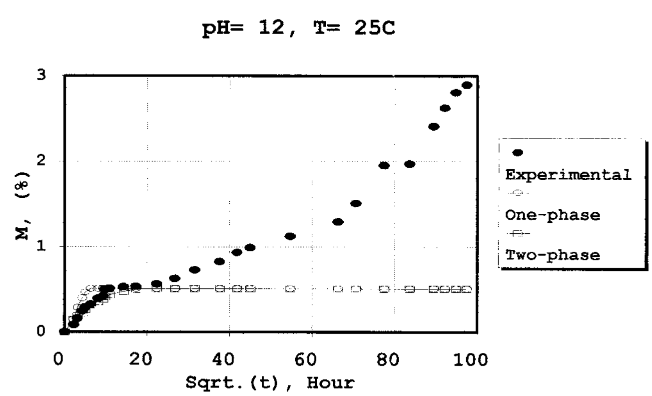

FIG. 6. M (%) versus $t^{1/2}$ for 10ARV Bar Specimen

J. Compos. Constr., 1999, 3(1): 12-19

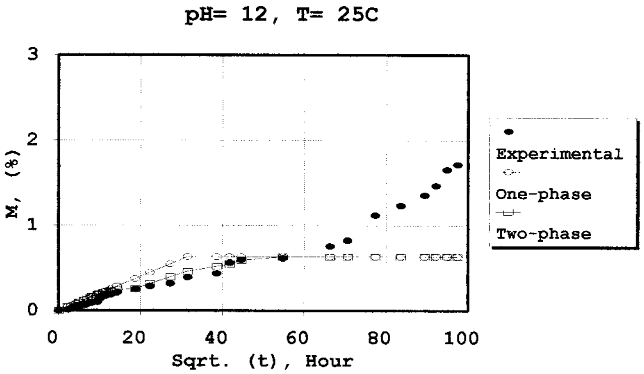

FIG. 7. M (%) versus $t^{1/2}$ for 20ARP Bar Specimen

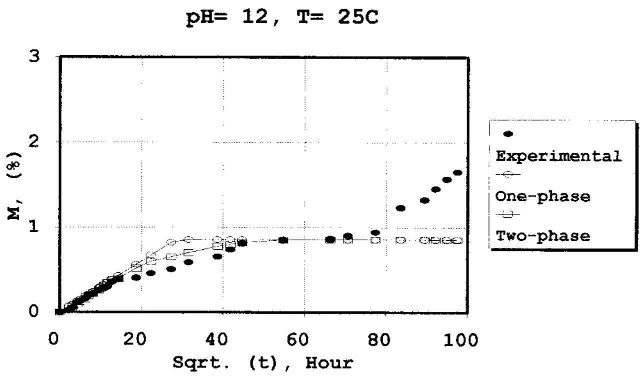

FIG. 8. M (%) versus $t^{1/2}$ for 20ARV Bar Specimen

3.1% for the 20ARV bar. For the 10ARP and 10ARV bars, the differences between the measured and predicted results were 7.2% and 3.4%, respectively. This indicates that Fick's law can be used to predict the loss of strength due to diffusion in an alkaline environment with a reasonable accuracy.

Alkaline Solution $Ca(OH)_2$: pH = 12; T = 60°C

M_m and D were higher than those in alkaline solution at room temperature. This indicates that temperature has a direct effect on the diffusivity. Similarly, the diffusivity of vinyl ester bars was lower than that of polyester bars. The highest percentage loss of tensile strength was observed in alkaline solution at 60°C. For the 10ARP bars, it was 28.0%, and for the 10ARV bars, it was 22.7%. As for the 20ARP and 20ARV bars, it was 16.9% and 14.1%, respectively. The lower measured percentage losses for the larger size bars was due to the fact that the penetrated area of these bars was proportionally smaller relative to the total cross section size of the bar. The percentage difference between the measured and predicted residual strengths was within 15% for the 10 mm (3/8 in.) diameter bars, and within 9% for the 19.5 mm (3/4 in.) diameter bars.

HCl Solution: pH = 3; T = 25°C

Similar trends in the behavior were observed in the M (%) versus $t^{1/2}$ curves for the acidic solutions. However, due to the space limitations, the corresponding figures are not shown here, but a discussion of the behavior is presented. Comprehensive data is provided by Tannous (1997).

In acidic solutions, M_m and D were higher for 10ARP and 20ARP than those for 10ARV and 20ARV bars. No noticeable loss in tensile strength was predicted in this case due to the very low concentration of chloride ions $[Cl^-]$ present in the solution. As a result, very small x values were predicted by (1). The predicted losses underestimated the actual losses in strength. This indicates the limitations of (1), which does not include the effect of moisture diffusion when no ions are present such as that from pure water. As was described earlier, pure water itself causes strength loss in bars. Therefore, additional studies must be conducted to develop new models or to extend the application of (1) to include the effect of water diffusion.

Seawater and Deicing Salt Solutions: T = 25°C

Generally, the diffusivity of polyester bars was higher than that of vinyl ester bars. However, less deviation in the measured M_m and D values was observed among the 10 mm (3/8 in.) and 19.5 mm (3/4 in.) diameter polyester and vinyl ester bars. Furthermore, very small differences were observed in the values of M_m between the 10 mm (3/8 in.) and the 19.5 mm (3/4 in.) diameter vinyl ester and polyester bars. Lower losses in tensile strength were observed in seawater than deicing salt solutions, which could be due to the lower chloride concentration and lower diffusivity. The difference between the measured and predicted residual strengths was within 15% in the case of 10 mm (3/8 in.) diameter polyester and vinyl ester rebars, and it was within 6% for the 19.5 mm (3/4 in.) diameter bars.

Ultraviolet Radiation

Only the 10 mm (3/8 in.) diameter AR glass/polyester and vinyl ester bars were exposed to ultraviolet radiation. The specimens were subjected to 31.7×10^{-6} J/s/cm^2 (2.04×10^{-4} J/s/sq in.) for a period of six months. Test results indicated that the rebars were not significantly affected by the ultraviolet radiation during the six months of exposure time. The average loss of tensile strength after exposure to UV light was 5.2% for the 10ARP bars and 5.7% for the 10ARV bars.

BEAMS REINFORCED WITH AR GLASS AND EXPOSED TO DEICING SALTS

Flexure tests conducted on concrete beams have shown that FRP bars can be used as effective reinforcement in concrete beams (Saadatmanesh and Ehsani 1991). Ten concrete beams measuring 200 mm (8 in.) $\times$ 405 mm (16 in.) in cross section and 2.44 m (8 ft) long were cast. Five beams were reinforced with two 10 mm (3/8 in.) ARP tension bars, and the remaining five beams were reinforced with two 10 mm (3/8 in.) ARV tension bars. Each beam was loaded in four-point bending as shown in Fig. 2. Table 5 summarizes the flexural test results of the beams. All beams failed as a result of rupture of tension bars since they were greatly underreinforced to insure this mode of failure. Therefore, the concrete in these beams was stressed relatively low, allowing the assumption of linear elastic analysis to calculate the tensile force in bars at failure from the flexural beam tests.

The beams were divided into three groups: (1) Two control specimens, B-1 and B-6, were kept dry and tested after one year; (2) four beams, B-2, B-4, B-7, and B-9, were tested after one year of exposure to deicing salt solutions; and (3) four beams, B-3, B-5, B-8, and B-10, were tested after two years of exposure to deicing salt solutions. Loads were applied in increments of 2 kN (450 lb) throughout the test, except near the failure, where the load increment was reduced to 1 kN (225 lb). The midspan deflection was recorded at the end of each load increment by a dial gauge with a 0.0254 mm (0.001 in.) resolution.

Fig. 9 shows flexural test results of beams B-1, B-2, and B-3 throughout the entire range of loading up to failure of the beam. The initial behavior was linear-elastic until the appearance of the first flexural crack. Beam B-1 was kept dry and tested in flexure to failure after one year. The percentage loss in tensile strength of reinforcement at failure was 7.3% com-

J. Compos. Constr., 1999, 3(1): 12-19

TABLE 5. Tensile Strength of 10 mm (3/8 in.) Diameter AR Glass/Polyester and Vinyl Ester FRP Rebars in Solutions and Concrete Beams

Environment/rebar type (1)	Initial P_0 (kN) (2)	P_r—six months (kN) (3)	Loss of strength (%) (4)	P_r—one year (kN) (5)	Loss of strength (%) (6)	P_r—two years (kN) (7)	Loss of strength (%) (8)
(*a*) 10 mm (3/8 in.) φ, AR glass/polyester							
Beam B-1	60.0	—	—	55.6	7.3	—	—
$Ca(OH)_2$ solution, T = 25°C	60.0	47.5	20.8	—	—	—	—
Beams B-2 and B-3	60.0	—	—	47.1	21.5	42.3	29.5
$NaCl + CaCl_2$ (2:1), 7% by weight	60.0	45.6	24.0	—	—	—	—
Beams B-4 and B-5	60.0	—	—	44.5	25.8	39.5	34.2
$NaCl + MgCl_2$ (2:1), 7% by weight	60.0	44.3	26.2	—	—	—	—
(*b*) 10 mm (3/8 in.) φ, AR glass/vinyl ester							
Beam B-6	61.3	—	—	52.6	14.2	—	—
$Ca(OH)_2$ solution, T = 25°C	61.3	53.5	12.8	—	—	—	—
Beams B-7 and B-8	61.3	—	—	45.8	25.3	42.3	31.0
$NaCl + CaCl_2$ (2:1), 7% by weight	61.3	47.4	22.7	—	—	—	—
Beams B-9 and B-10	61.3	—	—	46.2	24.6	44.0	28.2
$NaCl + MgCl_2$ (2:1), 7% by weight	61.3	45.9	25.1	—	—	—	—

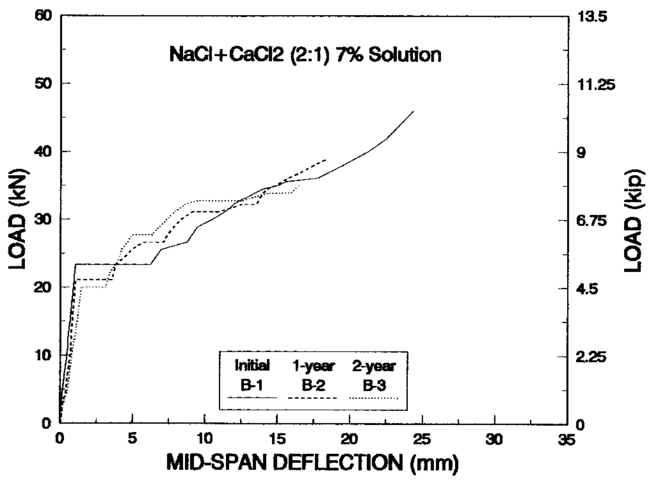

FIG. 9. Load versus Midspan Deflection for Beams B-1, B-2, and B-3

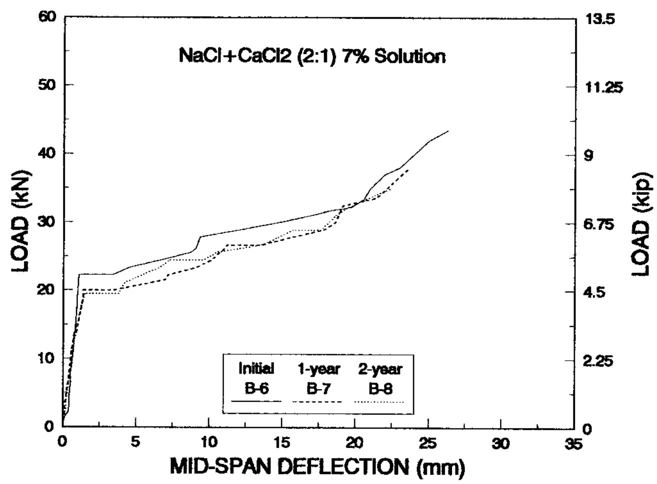

FIG. 11. Load versus Midspan Deflection for Beams B-6, B-7, and B-8

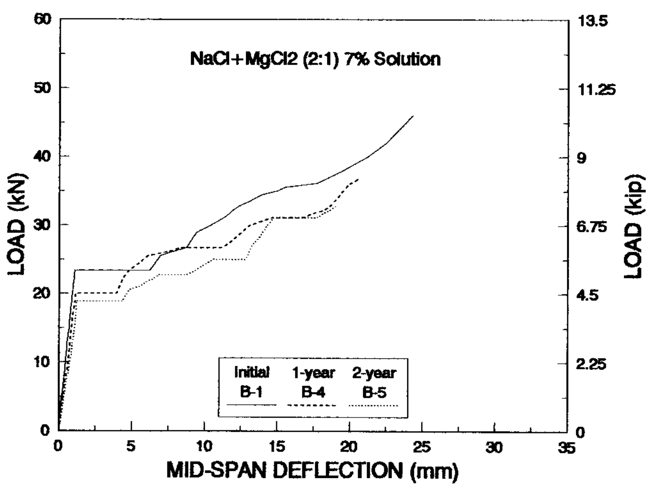

FIG. 10. Load versus Midspan Deflection for Beams B-1, B-4, and B-5

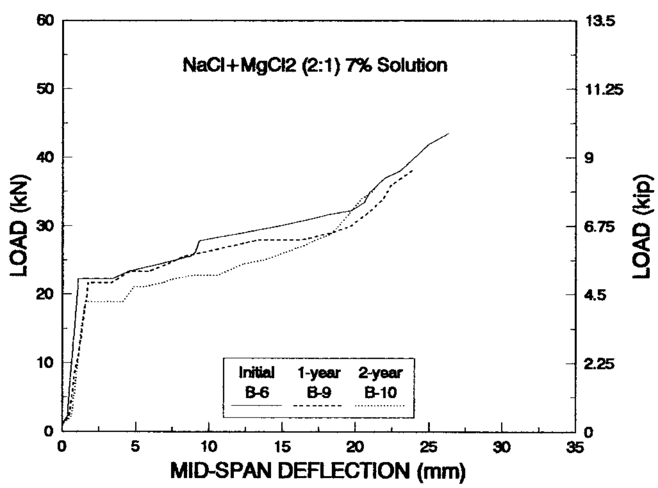

FIG. 12. Load versus Midspan Deflection for Beams B-6, B-9, and B-10

pared to 20.8% in accelerated tests after six months of direct exposure to alkaline solution. Beams B-2 and B-3 were placed in $NaCl + CaCl_2$ (2:1) 7% solution and tested after one year and two years, respectively. Test results showed 21.5% loss in tensile strength of polyester bars for B-2 and 29.5% for B-3. Similar observations were made about Beams B-4 and B-5 in Fig. 10. Furthermore, a loss in beam stiffness and bending capacity was also observed in all beams.

Figs. 11 and 12 show flexural test results of beams reinforced with two 10 mm (3/8 in.) diameter AR glass/vinyl ester bars in $NaCl + CaCl_2$ (2:1) 7% solution and $NaCl + MgCl_2$ (2:1) 7% solution, respectively. Beam B-6 was the control

J. Compos. Constr., 1999, 3(1): 12-19

beam and was kept dry and tested in flexure after one year. Initially, the beam behavior was linear elastic until the appearance of the first flexural crack, followed by a loss of stiffness after cracking. Again, as per design, all beams failed by rupturing of tension reinforcement. After one year exposure to concrete (i.e., B-6), the loss in tensile strength of AR glass FRP bars was 14.2%. However, in accelerated tests, where the bars were directly exposed to the alkaline solution simulating the hydrating cement, this loss was higher (20.8%). It was also observed that the loss in tensile strength of bars was not critically dependent on the type of deicing salt solution, but it depended more on the exposure time.

SUMMARY AND CONCLUSIONS

The diffusivity and moisture content at saturation was determined for four different AR glass FRP bar samples. Specimens of AR glass FRP bars were immersed in seven different solutions that resembled various conditions in the field. One-phase and two-phase diffusion models were used to predict the moisture diffusion and the residual strength of the bars. Ten beams reinforced with two 10 mm (3/8 in.) diameter AR glass FRP bars were tested in flexure until failure to examine the effects of exposure to deicing salt solutions. The following conclusions are drawn based on the results of the limited tests performed in this study:

1. Diffusion in FRP rebars is dependent on temperature and the type and concentration of the solution.
2. The vinyl ester used in this study exhibited lower diffusivity and provided a better protection to fibers against chemical attack than polyester.
3. The AR glass did not improve the behavior of bars made for this study in the alkaline environment of the concrete. Since the quality control during pultrusion makes a difference in the quality and durability of bars, it is suggested that AR glass bars made by other manufacturers also be tested to further quantify the durability of this type of bar in civil engineering environments.
4. Loss in strength was observed when AR glass FRP rebars were exposed to simulated marine environment.
5. Under most conditions, Fick's law together with the values of apparent diffusivity may be used to approximately predict the changes in moisture content and mechanical properties with time until the moisture content at saturation is reached.
6. The difference between the predicted residual strength and the measured residual strength was generally within 15% for 10 mm (3/8 in.) diameter polyester and vinyl ester bars, and within 10% for 19.5 mm (3/4 in.) diameter bars.
7. Fick's law generally applies to relatively short-term prediction of accelerated test results. At longer periods, due to microcracking and progressive matrix damage, the behavior becomes non-Fickian and cannot be predicted.

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J. Compos. Constr., 1999, 3(1): 12-19